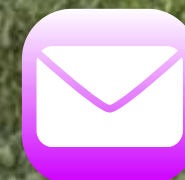
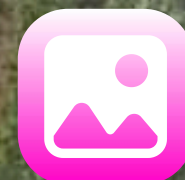
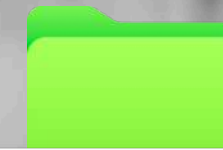
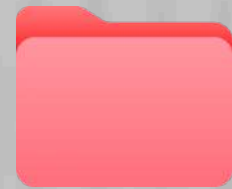


Crop Recommendation System Using Machine Learning: A Foundational Classifier Comparison

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FOR LEARNDEPTH
INTERNSHIP



**Problem:**

Classify whether environmental and soil conditions are suitable for selected crop categories.

Objective:

Build a foundational ML model that predicts the best crop from 7 soil/climate features.

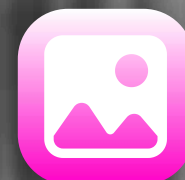
Inputs (7 features):

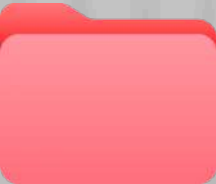
- N, P, K (soil nutrients)
- temperature, humidity, rainfall
- pH

Output: One of 22 crop categories



introduction





Crop Recommendation Dataset

Maximize agricultural yield by recommending appropriate crops

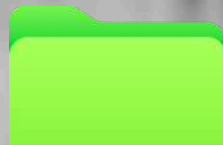
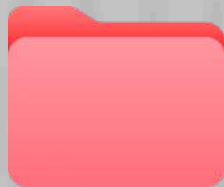
 [kaggle.com](https://www.kaggle.com/datasets/abhishek118/crop-recommendation-dataset)

Property	Detail
Total records	2,200 rows
Features	7 numeric
Classes	22 crops (100 samples each)
Missing values	None
Duplicates	None
Class balance	Perfectly balanced

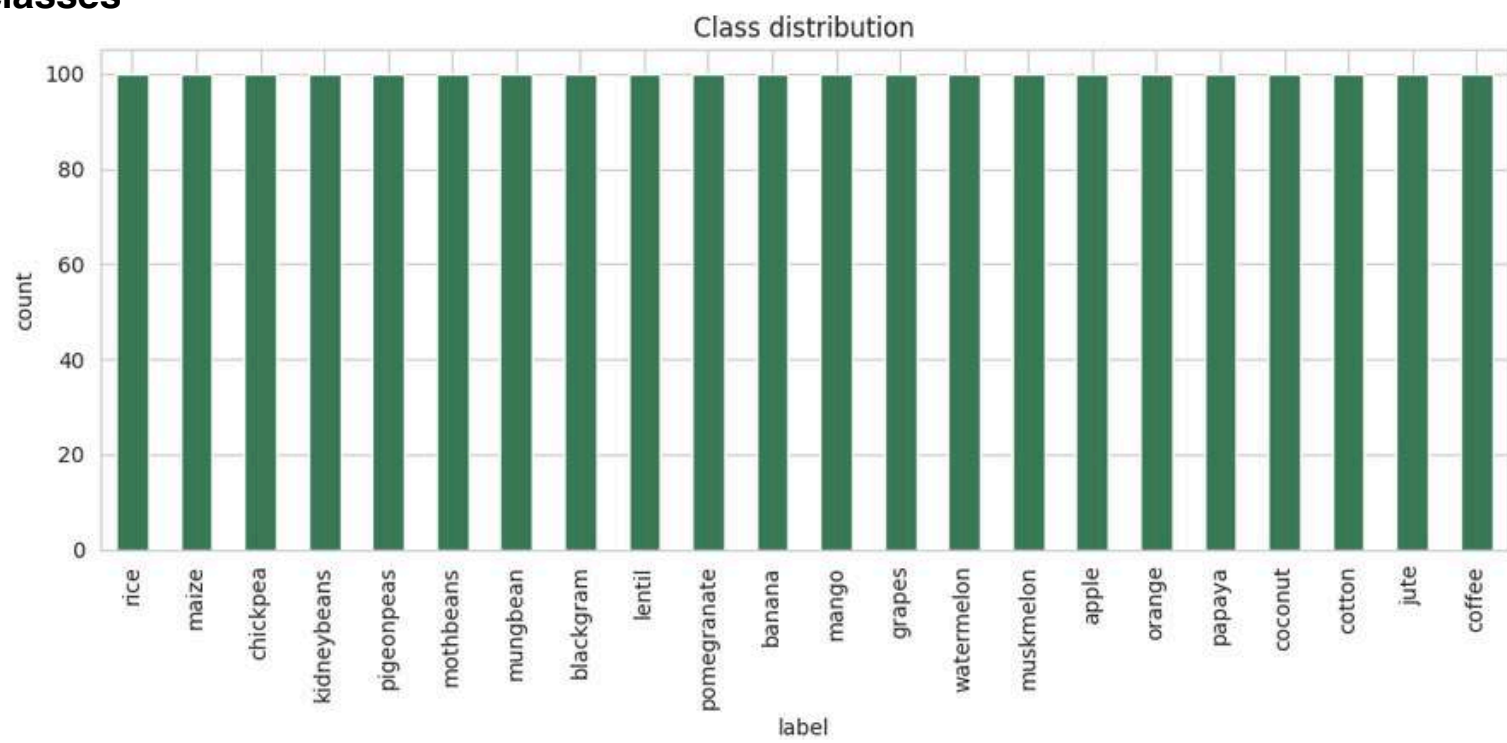


dataset

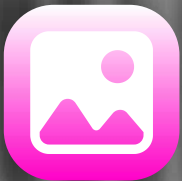




100 samples, 22 classes



dataset



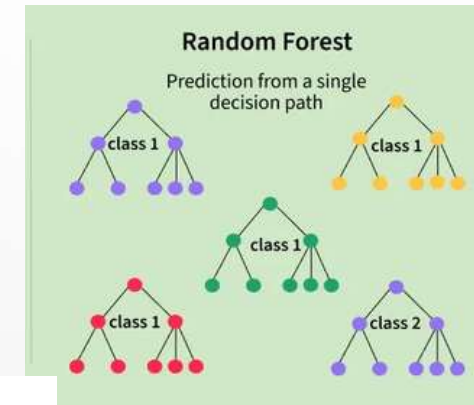
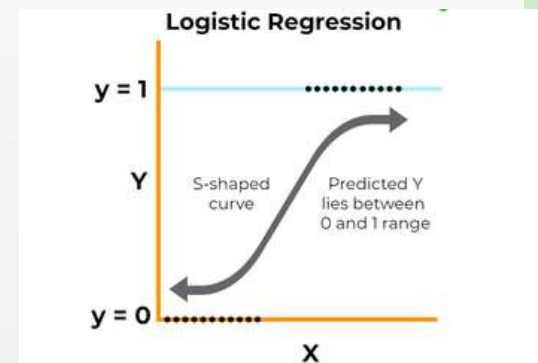
Load Data → EDA →
Preprocessing → Split
→ Train → Evaluate →
Interpret

Preprocessing:

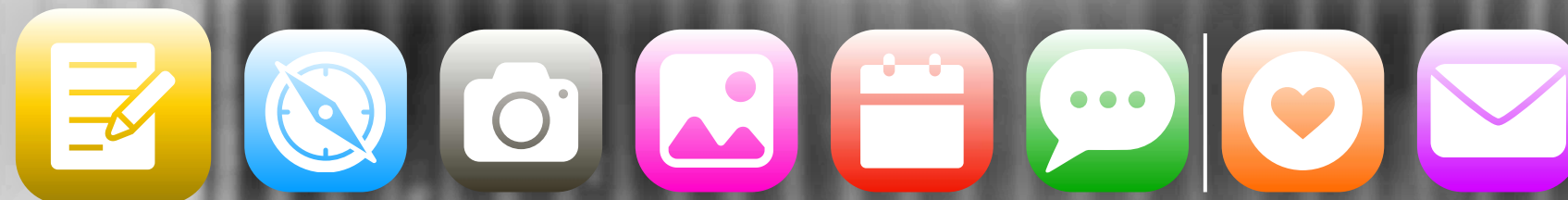
- Label encoding for target (22 crops)
- Train/test split 80/20 (stratified)
- StandardScaler (fit on train only)

Models:

- Baseline: Logistic Regression
- Main: Random Forest (300 trees)



methodology





Model	Accuracy	Macro-F1
Logistic Regression	97.27%	97.25%
Random Forest	99.32%	99.32%
RF 5-fold CV	99.50% ± 0.27%	—

Key findings:

- Random Forest outperforms LR by ~2%
- Cross-validation confirms model stability
- Only 3 misclassified cases out of 440



results





Rank	Feature	Importance
1	rainfall	0.222
2	humidity	0.212
3	K (potassium)	0.179
4	P (phosphorus)	0.154
5	N (nitrogen)	0.104
6	temperature	0.076
7	ph	0.054

Water (rainfall + humidity) is the primary filter; nutrients (N-P-K) follow.



features

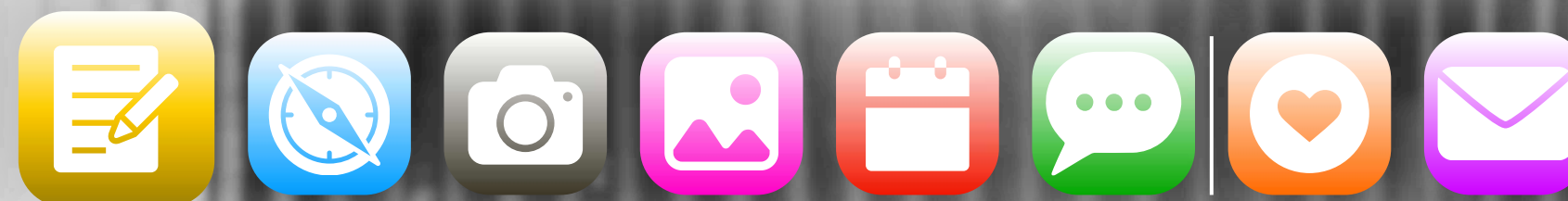


Misclassified: 3 / 440 (0.68%)

	true	pred
789	blackgram	maize
838	lentil	mothbeans
83	rice	jute

Errors occur between crops with genuinely overlapping profiles where model has learned meaningful boundaries. Its mistakes are **reasonable**.

error analysis





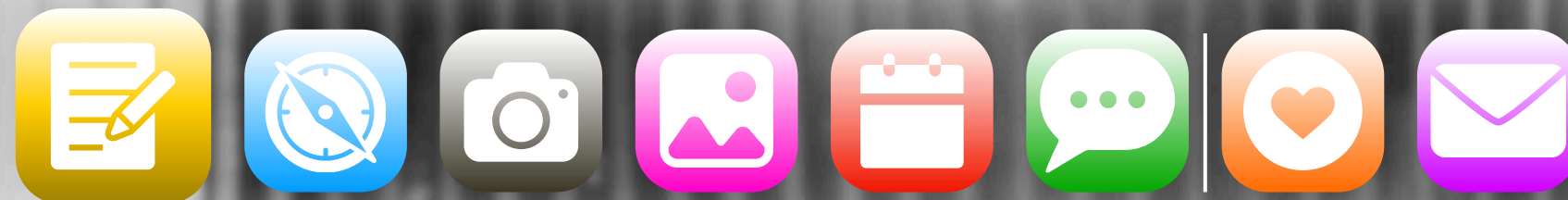
Data **limitations**:

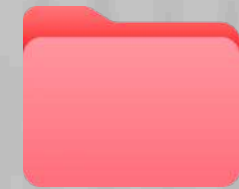
- Synthetic, not real farm data
- Perfectly balanced classes (unrealistic)
- No missing values or outliers
- No geographic/temporal context

Generalization **risks**:

- Real-world accuracy likely lower (70–90%)
 - Model will drift as climate/practices change
 - Needs validation on real data before deployment
- 

limitations





Summary:

- Random Forest achieves **99.32% accuracy** with stable cross-validation
- Rainfall, humidity, and K are the strongest predictors
- Errors are minimal and explainable

Critical caveat:

- This is a **proof-of-concept**, not production-ready
- High accuracy is due to synthetic dataset structure
- Real-world validation is the essential next step

Next steps:

1. Validate on real agricultural data
2. Add contextual features (soil type, season)
3. Try XGBoost/LightGBM
4. Build Streamlit app for demo



conclusion

